The Logarithmic Rearrangement Converse in Problem 28 Fails

Counterexample packet for arXiv:2004.03655

Automated mathematics run `fa_banach_001`, lane 8

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Status. Candidate counterexample, likely valid, pending human review. The result refutes the literal standalone equivalence in Problem 28. A scope warning concerning a possible implicit norm-growth hypothesis appears in Section 5.

1 Source problem

Astashkin and Milman ask in Problem 28 of *Extrapolation: Stories and Problems* [1] for a direct proof of the equivalence of the two rearrangement inequalities displayed as (10.8). The source is arXiv:2004.03655; the display occurs on page 41 and Problem 28 on page 42.

For a measurable function g on $(0,1)$, let g^* denote the decreasing rearrangement of $|g|$. Written with explicit constants, the two properties of an operator T are

$$\int_0^t (Tf)^*(s)\,ds \le C_1 \int_0^t f^*(s)\log\frac{t}{s}\,ds, \tag{A}$$

$$\int_0^t (Tf)^*(s)\log\frac{t}{s}\,ds \le C_2 \int_0^t f^*(s)\left(\log\frac{t}{s}\right)^2\,ds, \tag{B}$$

for every $f \in L^\infty(0,1)$ and every $0 < t \le 1$. The symbol $\preceq$ in the source means that the relevant constant is independent of f and t.

2 Idea of the proof

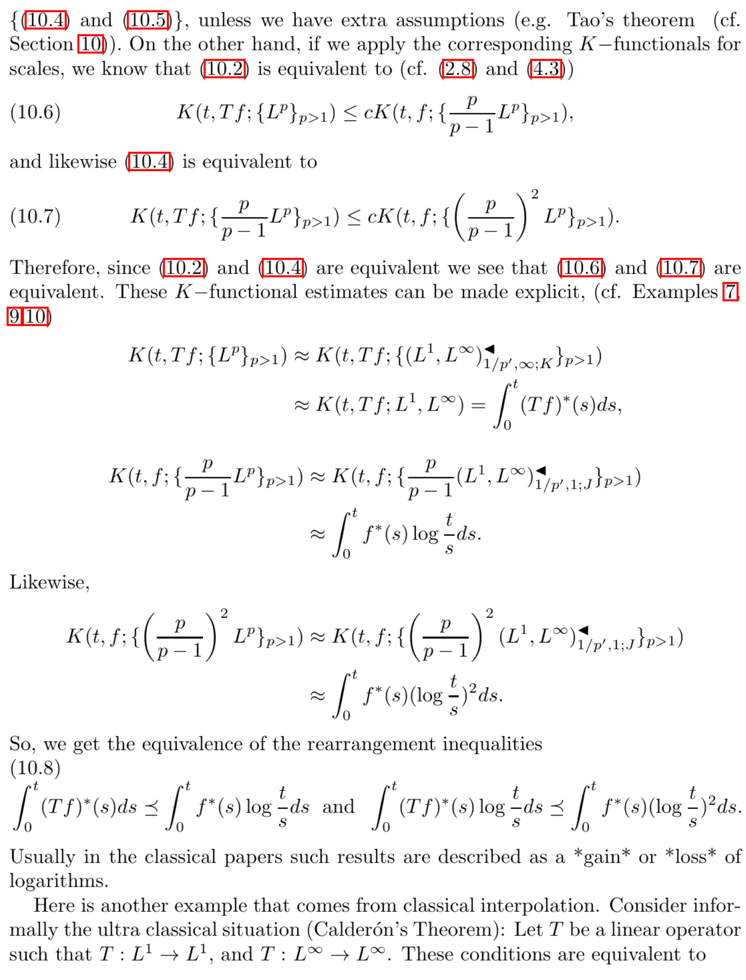
There is a genuine one-way relation: integrating (A) once in the scale variable adds one logarithm to each side, so (B) follows. The converse would have to remove a logarithm, and that operation is not order-preserving.

The obstruction can be isolated with a rank-one operator. Pair the input against the exact weight appearing on the right of (B), namely $h(s) = \log^2(1/s)$, and output the resulting scalar as a constant function. The dilation monotonicity of f^* then proves (B) with constant one. But indicators of increasingly short initial intervals make this functional larger than the right side of (A) by an unbounded logarithmic factor.

3 Result

Theorem 1. *The two properties* (A) *and* (B) *are not equivalent as standalone operator inequalities.*

1. *For every operator T for which the expressions are defined,* (A) *with constant C_1 implies* (B) *with constant $C_1/2$.*



{(10.4) and (10.5)}, unless we have extra assumptions (e.g. Tao's theorem (cf. Section 10)). On the other hand, if we apply the corresponding $K-$functionals for scales, we know that (10.2) is equivalent to (cf. (2.8) and (4.3))

$$K(t, Tf; \{L^p\}_{p>1}) \leq cK(t, f; \{\frac{p}{p-1}L^p\}_{p>1}), \tag{10.6}$$

and likewise (10.4) is equivalent to

$$K(t, Tf; \{\frac{p}{p-1}L^p\}_{p>1}) \leq cK(t, f; \{\left(\frac{p}{p-1}\right)^2 L^p\}_{p>1}). \tag{10.7}$$

Therefore, since (10.2) and (10.4) are equivalent we see that (10.6) and (10.7) are equivalent. These $K-$functional estimates can be made explicit, (cf. Examples 7, 9,10)

$$\begin{aligned} K(t, Tf; \{L^p\}_{p>1}) &\approx K(t, Tf; \{(L^1, L^\infty)^{\blacktriangleleft}_{1/p',\infty;K}\}_{p>1}) \\ &\approx K(t, Tf; L^1, L^\infty) = \int_0^t (Tf)^*(s)ds, \end{aligned}$$

$$\begin{aligned} K(t, f; \{\frac{p}{p-1}L^p\}_{p>1}) &\approx K(t, f; \{\frac{p}{p-1}(L^1, L^\infty)^{\blacktriangleleft}_{1/p',1;J}\}_{p>1}) \\ &\approx \int_0^t f^*(s)\log\frac{t}{s}ds. \end{aligned}$$

Likewise,

$$\begin{aligned} K(t, f; \{\left(\frac{p}{p-1}\right)^2 L^p\}_{p>1}) &\approx K(t, f; \{\left(\frac{p}{p-1}\right)^2 (L^1, L^\infty)^{\blacktriangleleft}_{1/p',1;J}\}_{p>1}) \\ &\approx \int_0^t f^*(s)(\log\frac{t}{s})^2 ds. \end{aligned}$$

So, we get the equivalence of the rearrangement inequalities

(10.8)

$$\int_0^t (Tf)^*(s)ds \preceq \int_0^t f^*(s)\log\frac{t}{s}ds \quad \text{and} \quad \int_0^t (Tf)^*(s)\log\frac{t}{s}ds \preceq \int_0^t f^*(s)(\log\frac{t}{s})^2 ds.$$

Usually in the classical papers such results are described as a *gain* or *loss* of logarithms.

Here is another example that comes from classical interpolation. Consider informally the ultra classical situation (Calderón's Theorem): Let T be a linear operator such that $T : L^1 \to L^1$, and $T : L^\infty \to L^\infty$. These conditions are equivalent to

Figure 1: The two inequalities (10.8) in the source paper, page 41.

2. *There is a rank-one linear operator T, bounded on $L^p(0,1)$ for every $1 < p < \infty$, which satisfies* (B) *with $C_2 = 1$ but satisfies* (A) *for no finite constant C_1.*

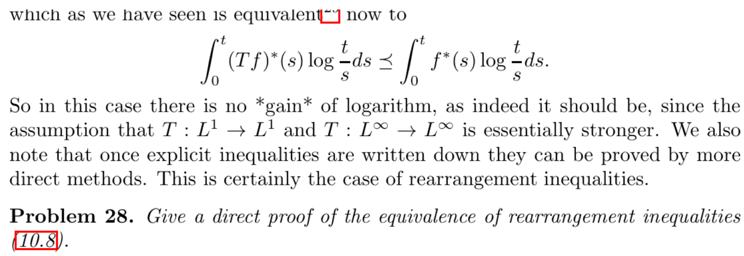
In particular, the literal equivalence requested in Problem 28 of [1] is false.

4 Proof

4.1 The valid implication

Assume (A). Integrate it with respect to du/u over $0 < u < t$. All integrands are nonnegative, so Tonelli's theorem gives

$$\begin{aligned} \int_0^t \frac{1}{u}\int_0^u (Tf)^*(s)\,ds\,du &= \int_0^t (Tf)^*(s)\log\frac{t}{s}\,ds, \\ \int_0^t \frac{1}{u}\int_0^u f^*(s)\log\frac{u}{s}\,ds\,du &= \int_0^t f^*(s)\int_s^t \log\frac{u}{s}\,\frac{du}{u}\,ds \\ &= \frac{1}{2}\int_0^t f^*(s)\left(\log\frac{t}{s}\right)^2 ds. \end{aligned}$$


which as we have seen is equivalent now to

$$\int_0^t (Tf)^*(s)\log\frac{t}{s}ds \preceq \int_0^t f^*(s)\log\frac{t}{s}ds.$$

So in this case there is no *gain* of logarithm, as indeed it should be, since the assumption that $T : L^1 \to L^1$ and $T : L^\infty \to L^\infty$ is essentially stronger. We also note that once explicit inequalities are written down they can be proved by more direct methods. This is certainly the case of rearrangement inequalities.

Problem 28. *Give a direct proof of the equivalence of rearrangement inequalities (10.8).*

Figure 2: Problem 28 in the source paper, page 42.

This proves (B) with $C_2 = C_1/2$.

4.2 Construction and boundedness

Put

$$h(s) = \left(\log\frac{1}{s}\right)^2, \qquad \ell(f) = \int_0^1 f(s)h(s)\,ds, \qquad Tf = \ell(f)\mathbf{1}.$$

This is a rank-one linear operator. If $1 < p < \infty$ and $q = p' = p/(p-1)$, then

$$\int_0^1 h(s)^q\,ds = \int_0^\infty x^{2q}e^{-x}\,dx = \Gamma(2q+1) < \infty.$$

Consequently Hölder's inequality yields

$$\|Tf\|_p = |\ell(f)| \le \|h\|_q\|f\|_p.$$

Thus T is bounded on every $L^p(0,1)$ with $1 < p < \infty$ (and it is also bounded on L^∞).

We shall use the elementary rearrangement estimate

$$|\ell(f)| \le \int_0^1 f^*(s)h(s)\,ds.$$

For completeness, h is nonnegative and decreasing. For every measurable $E \subset (0,1)$ of measure m, layer cake gives $\int_E h \le \int_0^m h$. Applying this inequality to the superlevel sets of $|f|$ and integrating over the level proves $\int |f|h \le \int f^*h$, which proves the estimate.

4.3 The second inequality holds

Because Tf is constant, $(Tf)^*(s) = |\ell(f)|$ on $(0,1)$. Therefore

$$\int_0^t (Tf)^*(s)\log\frac{t}{s}\,ds = t|\ell(f)|.$$

On the other hand, substitute $s = tu$ and use that f^* is decreasing. Since $0 < t \le 1$ implies $tu \le u$,

$$\begin{aligned}\int_0^t f^*(s)\left(\log\frac{t}{s}\right)^2 ds &= t\int_0^1 f^*(tu)\left(\log\frac{1}{u}\right)^2 du \\ &\ge t\int_0^1 f^*(u)h(u)\,du \\ &\ge t|\ell(f)|.\end{aligned}$$

Together with the preceding identity, this proves (B) with $C_2 = 1$ for every bounded f and every $0 < t \le 1$.

4.4 The first inequality fails

Let $0 < \varepsilon < 1$, set $f_\varepsilon = \mathbf{1}_{(0,\varepsilon)}$, and write $L = \log(1/\varepsilon)$. Direct integration gives

$$\ell(f_\varepsilon) = \int_0^\varepsilon \left(\log \frac{1}{s}\right)^2 ds = \varepsilon(L^2 + 2L + 2),$$
$$\int_0^1 f_\varepsilon^*(s) \log \frac{1}{s}\, ds = \int_0^\varepsilon \log \frac{1}{s}\, ds = \varepsilon(L+1).$$

At the permitted endpoint $t = 1$, the quotient of the left side of (A) by its right side without C_1 is therefore

$$\frac{\varepsilon(L^2 + 2L + 2)}{\varepsilon(L+1)} = L + 1 + \frac{1}{L+1} \longrightarrow \infty \qquad (\varepsilon \downarrow 0).$$

No finite C_1 can make (A) hold. This completes the proof of Theorem 1.

5 Verification and scope

The construction satisfies the explicit regularity present near the source problem: it is one fixed linear operator, it is bounded on every $L^p(0,1)$, $1 < p < \infty$, and all test functions used above belong to $L^1 \cap L^\infty$. The endpoint $t = 1$ is included in the source range.

There is nevertheless an important interpretive distinction. The discussion preceding (10.8) begins with the stronger scale estimate

$$\|T\|_{L^p \to L^p} \lesssim \frac{p}{p-1}.$$

If Problem 28 silently retains this as an additional hypothesis, then the example above is outside that narrower class. Indeed, the exact norm of the rank-one operator is

$$\|T\|_{L^p \to L^p} = \|h\|_{p'} = \Gamma(2p' + 1)^{1/p'} \asymp (p')^2 \qquad (p \downarrow 1),$$

by Stirling's formula. Thus the packet establishes a complete negative answer to the literal standalone equivalence of the displayed inequalities; it does not refute an explicitly augmented formulation with the $O(p')$ norm-growth assumption already imposed.

6 Novelty check and limitations

The run indexes contained no matching result or attempt for arXiv:2004.03655, Problem 28, logarithmic rearrangement inequalities, or rank-one counterexamples. Bounded web/arXiv searches on 9 August 2026 using the exact problem phrase, close variants, the source authors, and later Astashkin–Lykov–Milman work found no later answer or matching construction. This is not an exhaustive novelty claim. The result is noncomputational; human review should focus on the intended scope of "equivalence."

References

[1] S. Astashkin and M. Milman, *Extrapolation: Stories and Problems*, arXiv:2004.03655 (2020); Pure Appl. Funct. Anal. 6 (2021), no. 3, 651–707.